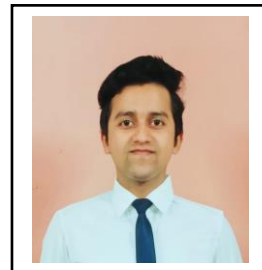




MAYANK RAJ

B.Tech – Production Engineering
Gender: Male
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Educational Qualification

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2022-Present	B.Tech- Production	NIT, Trichy	8.06
2021	Class XII	Loyola School, Jamshedpur, ISC	87.60%
2019	Class X	Loyola School, Jamshedpur, ICSE	90.20%

Academic Achievements

- Recipient of the prestigious **Tata Steel Millennium Scholarship** in recognition of consistent academic excellence.
- Achieved **first and second place** at zonal level in the **National Science Olympiad (NSO)** and **International Mathematics Olympiad (IMO)**, conducted by SOF, in 2017.
- Pursuing a minor in **computer science** alongside a major in production engineering at Nit Trichy.

Internship Experience

- **Tata Steel | Data Analyst Intern:** May 2025 - July 2025
Developed and evaluated machine learning models including Linear Regression, Random Forest, XGBoost, and KNN to predict sulphur removal efficiency, achieving an R^2 score of up to 0.88. Engineered domain-specific features such as the Mg-to-Lime ratio and applied Particle Swarm Optimisation (PSO) to enhance ensemble model performance and support process optimisation at LD3.
- **FMS Delhi | Research Intern:** Dec 2024 - Jan 2025
Updated an academic sentiment analysis work using recent Twitter and product review data related to smartphones. Automated tweet extraction and preprocessing with Tweepy, followed by sentiment classification using custom Python logic in Google Colab. Replaced outdated plots with new insights and visualizations using Seaborn and Matplotlib for models like OnePlus 12R and Samsung S24 Ultra.

Other Projects

- **Olympics Data Analysis | Self Project:** Oct 2024 - Dec 2024
Developed an interactive Olympics data analysis web app using Python, Stream lit, and Plotly to explore 120+ years of data. The app includes dynamic filters for year, country, and sport, with modules for medal tallies, athlete statistics, and participation trends. Data was preprocessed using Pandas and organized through modular helper functions. Insights were visualized using heatmaps, line charts, and distribution plots for clear and engaging analysis.

- Inventory Optimization | Self Project:** May 2025 – June 2025
 Built a SQL-based inventory monitoring system to reduce stockouts, overstocking, and inefficiencies across 5,000+ SKUs. Developed advanced queries for stock levels, reorder points, and supplier performance. Normalized data designed an ERD and created KPI dashboards to improve store and warehouse visibility.
- Movie Recommender System | Self Project:** Mar 2025 – Apr 2025
 Built a content-based movie recommender using cosine **similarity** on 5K+ titles. Integrated **TMDB API** for **poster retrieval** and developed a responsive Streamlit UI for **real-time top 5** suggestions. Optimised performance with **Pickle-based model serialisation** for faster load times.

Technical Skills and Certifications

- Languages:** Java, Python
- Libraries/Frameworks:** NumPy, Pandas, Scikit-learn, Matplotlib
- Database:** MySQL
- Other Software:** MS Excel, Tableau, Visual Studio, Git, Canva, Figma
- Certifications:** Summer Analytics IITG, Time Series Analysis IITG

Positions of Responsibility

- Manager at the Entrepreneurship Cell, NIT Trichy:** Aug 2023 – May 2025
 Contributed to the D2C Alum Startup's e-commerce team, leading cardamom trademark registration and refining product details. Worked with the E-Summit'24 marketing team, securing sponsorships through LinkedIn and cold emails, and assisted in organizing 'Breakthrough-Pitch Perfect' for HeadStart'24.
- Marketing Manager, Prodigy'24:** Feb 2024 – May 2025
 Tracked sponsors for Prodigy'24 (annual symposium of production department), secured partnerships with 6 companies contributing over 40% of the total collection, trained juniors on pitching and branding, and contributed to event planning.
- Publicity Coordinator at Pragyan'24:** July 2023 – June 2024
 Directed 10,000+ footfall from 400+ colleges nationwide by creating innovative offline and online publicity campaigns, focusing on metropolitan areas. Managed Pragyan's social media, achieving 80,000+ impressions and 20% growth in one year.
- Web-Ops Manager at Taskforce:** Aug 2023 – May 2025
 Contributed to UX design and front-end development for the website of Taskforce, the social initiative hub of NIT Trichy.

Extracurricular Activities

Social Activities:

- Contributed to the **Blood Donation Camp** by Jamshedpur Blood Centre through voluntary blood donation, supporting community health.
- Served as a volunteer under the **NSS program**, NIT Trichy chapter, focusing on providing individual attention and education to students in rural areas.

Sport Activities:

- Secured a **gold medal** at Inter-School 400m relay race, among **250+ participants** (2018).
- Secured Second Place in **Badminton (Singles)** in **Aaveg'23** - the inter-hostel cultural fest organised by NIT Trichy.

Cultural Activities:

- Secured first place in **Choreonite in Aaveg'23**- the inter-hostel cultural fest organised by NIT Trichy.